

A Comparative Analysis of XGBoost and SVM in Classifying Internet Gaming Disorder Severity

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Abstract— Internet Gaming Disorder (IGD) has become a serious public health problem. However, the prediction of severe IGD is still challenging because traditional clinical assessments mainly depend on subjective diagnosis and manual evaluation. In this paper, we present an automated computational pipeline for classifying severity levels of IGD using Extreme Gradient Boosting (XGBoost) and Support Vector Machine (SVM) architectures on a multidimensional behavioral and psychological data set. The initial baseline models had data leakage via a tautological proxy feature `loss_of_other_interests`, which led to artificially high accuracy metrics above 0.92. The systematic ablation of this feature followed by rigorous hyperparameter optimization using a Stratified 3-fold GridSearchCV achieved a Macro F1-Score of 0.7326 for XGBoost, better than the Macro F1-Score of 0.7218 of the SVM. While SVM achieved optimization of the linear classification margins, it was susceptible to high-magnitude exposure measures such as the daily gaming hours dominating the feature set. In contrast, XGBoost sequential boosting was able to capture subtle non-linear interactions of active functional impairments and psychological symptoms without memorizing the majority class. These results confirm the impossibility of a single pathognomonic shortcut to describe the severity of IGD and provide a mathematically robust baseline for early detection and intervention protocols in school and healthcare settings.

Keywords—Internet Gaming Disorder, Extreme Gradient Boosting, Support Vector Machine, Addiction Risk Classification, Hyperparameter Optimization

I. INTRODUCTION

Video games are among the most popular forms of entertainment in the world today. Modern gaming is no longer tied just to consoles; the explosion of mobile platforms and e-sports has made digital worlds accessible around the clock. Unsurprisingly, the surge in popularity has brought Internet Gaming Disorder (IGD) to the forefront as a serious threat to public health. All empirical data from around the world points to a rapidly growing number of people from all demographics affected by this disorder. It is often associated with serious emotional problems. People don't play for fun anymore. A lot of people use it as an escape mechanism that runs out of control very quickly. Unrestrained gaming behavior can often lead to depression, chronic anxiety, and a significant deterioration of the general health of players [1]. These

symptoms deteriorate over time and subtly, so we need to find better ways to analyze these addiction indicators to catch the problem early on [2].

How addicted is the user. That's a very specific question. Many parents and doctors obsess about the clock but raw screen time is not the be-all and end-all and clinical evidence shows play hours rarely tell the whole story. Some people can play for hours and not neglect their real life and some people can break down after a few hours of intense, compulsive play. Disturbance in sleep pattern [3] will usually first affect the internal system of the body, digital dependence. The first red flag is almost always a physical one. There are many things to consider when understanding how severe the user's addiction is. These include the number of hours the user slept, emotional tolerance, and the user's perception of sleep quality. This is particularly true for college students who are under a lot of academic pressure [4]. Without the right coping mechanisms they are very vulnerable to being caught up in this destructive cycle.

So far, the dynamics have been addressed mostly by researchers and clinicians in conventional descriptive approaches. Current methods still depend largely on subjective criteria for diagnosis, manual psychometric testing and simple statistical modelling. The issue with manual evaluations is that they are slow and reactive. While the patient's habits may have deteriorated, a psychologist might have finished scoring a psychometric test. These observational studies have taught us a lot about the relationship between interactive media use and mental health decline, however there is a large gap in the real time application of this data. We know the symptoms in theory, but we have difficulty in following them in practice.

Our mental states and physical habits interact in messy, non-linear ways that traditional analytic frameworks have a hard time capturing. A standard statistical model simply can't separate out the effect of a decline in sleep quality today on a player's mood or gaming intensity tomorrow. Designing early intervention protocols is very difficult as the field still depends on fragmented and subjective human cut-offs instead of hard computational thresholds. Schools and hospitals have to guess at when a user goes from being a heavy gamer to a severe addict. Today, there is no standard that can take multi-dimensional survey data and automatically provide a clear

classification of addiction risk. This is a major blind spot in modern psychiatry and education care.

We undertook this study to address that gap. We are moving from manual clinical assessments towards proactive, data-driven prediction. Hence, we used state-of-the-art machine learning methods, namely Extreme Gradient Boosting (XGBoost) and Support Vector Machine (SVM), to classify IGD severity. We aim to develop an automatic and accurate classification framework, by exploring different features such as mood states, daily hours of game play and sleep duration. The goal is to provide health care professionals and educational institutions with a reliable algorithmic basis for early intervention.

II. RELATED WORKS

Diagnosing Internet Gaming Disorder (IGD) is very difficult. Thus, recently researchers have started to move away from standard psychometric tests, and toward Machine Learning (ML) classifiers. This represents a promising research domain, as the application of ML to IGD remains a novel concept. Fortunately, we have a lot of good computational research on its parent categories: general Internet and Smartphone Addiction. These addictions are very similar in psychological markers and tabular data structure which makes predicting them a great benchmark to aim for. Most early efforts in this area were limited to simple algorithms. One study examined smartphone addiction among college students [5]. Using only the standard Support Vector Machine (SVM) models, they could reach 81.3% accuracy. Another project predicted the risk of internet addiction using Logistic Regression [6] and achieved a macro F1-Score of 84.0%. The take-home message from these numbers is that basic models can capture general addiction trends, but they often fail to map the complex, non-linear behaviors of users, without advanced feature optimization.

Finally, the emphasis was put on more complex setups for better results from complex human data. One study [7] on problematic smartphone use achieved a diagnostic accuracy of 82.59% using behavioral log data with Random Forest, an ensemble algorithm. A comparable finding was reported in a recent neuroimaging study using fMRI data [8]. They used SVM for classification of heavy smartphone users. They used recursive feature elimination method to discard redundant data and their SVM model performed well with an accuracy of 79.9% and F1-Score of 70.8%. The results show that SVMs can be very useful in discriminating complex behavioral datasets.

However, the challenge of effectively predicting the severity level of IGD with multidimensional tabular data still remains. The main problem is the imbalance in classes. In any dataset, the cases that are severe addiction cases are usually the minority. This completely screws up standard models. Luckily, recent tests with ensemble methods [9] indicate that Gradient Boosting frameworks can handle this rather well. The data of internet use was imbalanced but they could handle it and still got out with a decent 69.7% macro F1-Score. Another project [10] attempted to cluster users according to problematic smartphone habits. They found that tree based architectures such as Random Forest are good at processing complex psychological surveys and demographic details with a combined classification accuracy of 73.0%. Then we benchmark XGBoost and SVM on these established benchmarks. We want to see if we can do better than that F1-

Score and create a better automated tool to classify IGD severity.

III. METHODOLOGY

The methodological pipeline used for this study is shown in Figure 1.

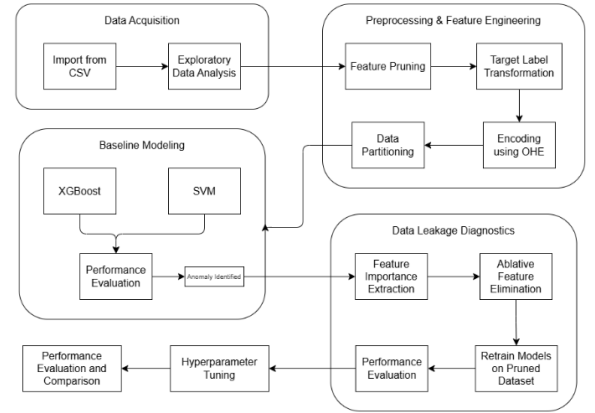


Figure 1. Flowchart of Research Methodology

A. Data Acquisition and Dataset Characteristics

The study used a quantitative experimental design with secondary data from the ‘Gaming and Mental Health’ repository, a public data set available at Mendeley Data [11]. From the first inspection we can see that there are a lot of missing values in the `grades_gpa` and `work_productivity_score` columns. Hence, as a result, these two feature columns were removed from the dataset to keep the sample volume without any imputation bias. This is consistent with the practice in the analysis of clinical surveys of discarding incomplete variables to avoid adding synthetic noise to predictive models [12]. In addition to the missing values, other features were removed based on the Pearson correlation matrix derived from initial exploratory data analysis. The main features used in the modelling process are listed in Table I.

TABLE I. SUMMARY OF CORE FEATURES AND TARGET LABEL

Feature Name	Data Type	Operational Description
age	Numeric	Continuous value representing the age of the participant in years
gender	Categorical	Demographic indicator representing the self-identified gender of the participant.
daily_gaming_hours	Numeric	Continuous value representing the average hours spent gaming daily
withdrawal_symptoms	Categorical	Binary indicator of the presence of withdrawals when not playing games
continued_despite_problems	Categorical	Binary indicator representing the participant continuing to game despite negative effects
gaming_addiction_risk_level	Categorical	Multi-class target variable mapped as 0 to 3.

B. Data Preprocessing and Feature Engineering

Before being used for model training, the dataset went through four distinct preprocessing steps, beginning with feature pruning.

The first features to be removed were those with missing values, namely `grades_gpa` and `work_productivity_score`. Following that, non-informative features were also eliminated, such as unique identification variables like `record_id` that have zero relevance to the target variable. Then, features with excessive cardinality or non-contributive variance, including `primary_game` and `gaming_platform`, were dropped to improved computational efficiency and mitigate overfitting. The pearson correlation matrix also revealed severe redundancy and multicollinearity between three features: `daily_gaming_hours`, `social_isolation_score`, and `face_to_face_social_hours_weekly`, thus the latter two features were removed from the dataset to prevent predictive inflation.

After the feature pruning, the next step is the conversion of the categorical ordinal target variable for addiction severity (`gaming_addiction_risk_level`) into a numerical scale. The labels were mapped manually with Low mapped to 0, Moderate mapped to 1, High mapped to 2 and Severe mapped to 3.

While the target variable was mapped manually, the other categorical predictors capturing demographic, behavioral, and psychological profiles, namely `gender`, `game_genre`, `sleep_quality`, `academic_work_performance`, `mood_state`, and `mood_swing_frequency`, were processed via One-Hot Encoding (OHE). The encoder was instantiated with the setting `handle_unknown='ignore'` and `sparse_output=False` to generate dense matrices for readability.

The post-engineered feature space was then divided 80:20. This division was also performed with `stratify=y` and `random_state=42` so that the proportions of classes in both data subsets are balanced and reproducible.

C. Predictive Modeling and Data Leakage Detection

This study uses two different classification models for the automatic classification of Internet Gaming Disorder (IGD) risk levels and compares the two models:

The first model is Extreme Gradient Boosting (XGBoost) which is an ensemble model based on trees with objective function specified as `multi:softmax` for multi-class classification and the performance metric during model training is the multi-class logarithmic loss (`eval_metric=mlogloss`). Since our data is imbalanced, we compute a custom sample-weighting vector using `compute_class_weight` (`class_weight='balanced'`) which we pass directly into the `sample_weight` argument of our model. XGBoost was selected for its regularization feature and ability to handle complex tabular data without overfitting [12].

The second model is Support Vector Machine (SVM). Before training, we standardize the feature inputs to zero mean and unit variance with `StandardScaler`. We then trained the SVM model with a balanced `class_weight` parameter (`class_weight='balanced'`) to address the class imbalance and allow a model that can better map the non-linear classification boundaries prevalent in psychological data [13].

To confirm the validity of the used predictors, the two models were trained and evaluated on the preprocessed feature

space and the near-perfect performance metrics strongly indicate data leakage.

To determine the variables responsible for the inflated prediction scores, the feature importance measures were extracted directly from the training split using the native relative gain as calculated by the XGBoost model. We used XGBoost to assess feature importance, since its gain metrics explicitly follow how features determine the splits in the trees, rather than the complicated mathematical transformations of hyperplane coefficients in the non-linear SVM.

Feature ablation of the features with the highest relative gains was performed to identify the feature responsible for the data leakage and was removed from the feature space. The models were then evaluated on this new dataset to check if the data leakage had been fixed.

B. Hyperparameter Tuning

The models were retrained after fixing data leakage, with hyperparameter tuning using `GridSearchCV` with a stratified 3-fold cross-validation. The hyper-parameter settings tested for both XGBoost and SVM are shown in Table II.

TABLE II. HYPERPARAMETER SEARCH GRID

Model	Hyperparameter	Evaluated Grid
XGBoost	Estimators	[100, 200, 300]
	Learning Rate	[0.05, 0.1, 0.2]
	Max Depth	[3, 5, 7]
SVM	Penalty Parameter	[0.1, 1, 10]
	Kernel Coefficient	[scale, auto]
	Kernel Type	[rbf, linear]

D. Performance Evaluation Metrics

The performance of the model was evaluated using four metrics namely Accuracy, Precision, Recall and F1-Score. Eq. 1 provides the global accuracy of the model, while the equations shown in Eq. 2, Eq. 3, and Eq. 4 provide the per-class baselines. In these equations, TPC, TNC, FPC and FNC denote True Positives, True Negatives, False Positives and False Negatives for a given class c .

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad \text{Eq. 1}$$

$$\text{Precision}_c = \frac{TP_c}{TP_c+FP_c} \quad \text{Eq. 2}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c+FN_c} \quad \text{Eq. 3}$$

$$\text{F1-Score}_c = 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad \text{Eq. 4}$$

Accuracy was only used to gauge data leakage as it alone cannot be a reliable measure of the model performance given the substantial class imbalances across addiction risk levels. Therefore, the macro-averaged metrics were used, defined by Eq. 5, Eq. 6, and Eq. 7.

$$\text{Macro Precision} = \frac{1}{N} \sum_{c=1}^N \text{Precision}_c \quad \text{Eq. 5}$$

$$\text{Macro Recall} = \frac{1}{N} \sum_{c=1}^N \text{Recall}_c \quad \text{Eq. 6}$$

$$\text{Macro F1-Score} = \frac{1}{N} \sum_{c=1}^N \text{F1-Score} \quad \text{Eq. 7}$$

The primary metric used for objective performance evaluation is the macro-averaged F1-Score, representing the harmonic mean of precision and recall accounting for skewed class distribution [14].

IV. RESULTS AND DISCUSSION

A. Dataset Inspection and Preprocessing

To check for multicollinearity, a Pearson correlation matrix was plotted across all numerical features and the target variable as shown in Figure 2. Several variables were found to be highly multicollinear ($r > 0.85$) with others, in particular `daily_gaming_hours`, `face_to_face_social_hours_weekly` and `social_isolation_score`. These features had correlations with the target label of $r = 0.81$, $r = -0.73$, and $r = 0.74$, respectively. Finally, `face_to_face_social_hours_weekly` and `social_isolation_score` were dropped from the final feature set to avoid model instability but `daily_gaming_hours` was retained since it was more correlated with the target.

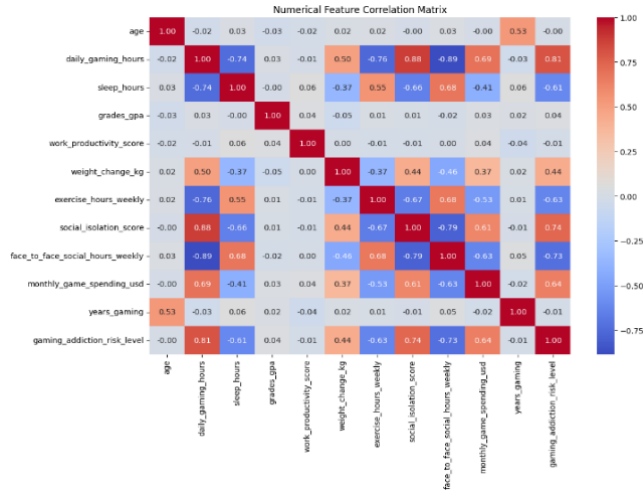


Figure 2. Pearson Correlation Matrix for Numerical Features

B. Data Leakage Detection Results

When the first time the initial feature space models are tested and the classification metrics obtained by XGBoost and SVM are near perfect. Table III shows the accuracy values for both models above 0.92. Such performance in behavior modeling is normally an indication of data leakage, not a real predictive power [15].

TABLE III. INITIAL MODEL PERFORMANCE

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
XGBoost	0.9800	0.9681	0.9692	0.9681
SVM	0.9200	0.8813	0.8850	0.8813

Feature importance scores natively computed by XGBoost highlighted four primary predictors: `daily_gaming_hours`, `loss_of_other_interests`, `continued_despite_problems`, and `withdrawal_symptoms`, shown in Figure 3.

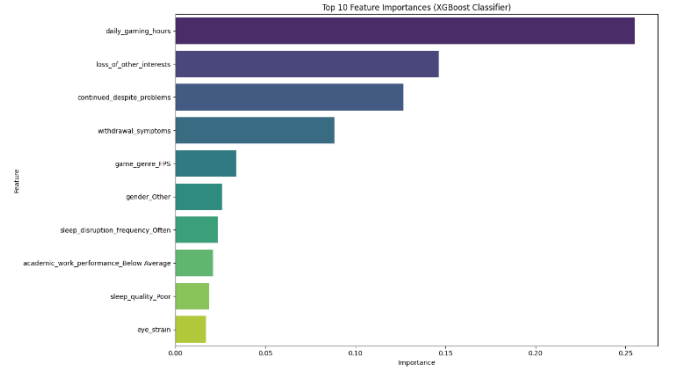


Figure 3. Top 10 Feature Importances in XGBoost Classifier

The `daily_gaming_hours` had the largest individual relative gain but feature ablation revealed that `loss_of_other_interests` was critical to the multi-class classification boundary. As shown in Table IV, when we removed this one feature, the model accuracy range was reduced to a more realistic 0.75-0.76. This confirms that `loss_of_other_interests` was the primary driver of data leakage.

TABLE IV. MODEL PERFORMANCE ON PRUNED FEATURE SET

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
XGBoost	0.7550	0.6771	0.6846	0.6805
SVM	0.7600	0.6889	0.6980	0.6925

Based on the evaluation of the optimized feature space, the SVM performed marginally better compared to XGBoost. As detailed in Table IV, the SVM achieved a Macro F1-Score of 0.6925, slightly edging out XGBoost's Macro F1-Score of 0.6805.

C. Hyperparameter Tuning Results

To find the performance ceiling for both models, we carried out hyperparameter tuning using a stratified 3-fold GridSearchCV on both models. Following the grid search, the optimal hyperparameters for XGBoost converged at 100 estimators, learning rate of 0.05, and `max_depth` of 3, while the SVM framework stabilized at `C=1`, `gamma='scale'`, and `kernel='linear'`. The final performance results of these hyperparameters are shown in Table V.

TABLE V. MODEL PERFORMANCE AFTER HYPERPARAMETER TUNING

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
XGBoost	0.7850	0.7242	0.7593	0.7326
SVM	0.7750	0.7176	0.7292	0.7218

Through the hyperparameter tuning, XGBoost was able to achieve a Macro F1-Score of 0.7326, and the SVM achieved a Macro F1-Score 0.7218.

D. Discussion

The better performance of the linear kernel over the non-linear Radial Basis Function (RBF) kernel on the SVM shows that the multi-class decision boundaries that remained were mostly linear after fixing the data leakage. The regularization parameter was fixed to $C = 1$, which was found to be optimal

with respect to a balanced margin penalty - thus allowing for small training errors to improve generalization performance.

The best parameters for XGBoost were found to be `max_depth = 3` and `learning_rate = 0.05`. The depth of the tree was constrained over 100 iterations to prevent overfitting to noise in the survey data. This resulted in a more general model that reduced the domination of the majority class, leading to a better Macro Recall of 0.7593.

From then on, it can be said that although XGBoost obtained a higher Macro F1-Score than SVM, its real advantage is demonstrated by its Macro Recall of 0.7593, against SVM's Macro Recall of 0.7292. This confirms that XGBoost was much more successful in correctly identifying the minority class, and thus it was the better model for this dataset.

Clinically, a Macro F1-Score of 0.72–0.73 is indicative of a strong model for an automated diagnostic screener, especially considering the subjective noise present in psychological self-reported data. The comparable performance of both modeling frameworks suggests that the severity of internet gaming disorder cannot be captured by a single diagnostic shortcut but rather depends on a combination of concurrent indicators where quantitative exposure metrics are considered together with functional impairments. The scores demonstrate that the models are able to identify high-risk individuals without falling back on the majority class, as the macro-averaging method gives equal weight to the small, high-severity classes and to the baseline population.

V. CONCLUSION

In this study, we developed an automated computational framework for the classification of the severity of Internet Gaming Disorder (IGD). Instead of traditional subjective surveys and manual assessment, we evaluated two machine learning architectures, XGBoost and SVM. One feature (`loss_of_other_interests`) in the first baseline evaluations was found to have a big data leakage problem. By systematically removing this feature and tuning the hyperparameters rigorously, the models produced more realistic and mathematically honest metrics. XGBoost delivered a better performance than SVM, with a maximum Macro F1-Score of 0.7326 in contrast to 0.7218. This is an improvement because the sequential boosting in XGBoost prevents the model from simply memorizing the majority class. It also dealt better with dominant features than the fixed linear boundaries of the SVM. This allowed the algorithm to detect the subtle non-linear interactions within the psychological data.

One of the main findings of this study is that there is no reliable single isolated marker for diagnosing gaming addiction. But it's not just about the number of days, it's a delicate balance of things happening all at once, how long a person has been exposed, how much their day-to-day life has been disrupted, to create an accurate risk profile. We note that a macro F1 score in the low 0.70s may seem modest at first glance, but is a very strong and realistic baseline, given the noisy nature of complex and subjective human behavioral data.

Future work should evaluate these models on longitudinal datasets to understand how gaming habits evolve over time. In addition, the integration of physiological sensor data into the predictive pipeline in real-time may help to further reduce human reporting bias and can dramatically improve the

accuracy of early intervention tools. adding real-time physiological sensor data to predictive pipeline very helpful also. This can further reduce human reporting bias and dramatically improve the accuracy of these early intervention tools.

DATA AVAILABILITY

The dataset used is publicly available on the Mendeley Data repository titled "Gaming and Mental Health" published by Gagan Randhawa on March 18, 2026 [11] doi: 10.17632/gnmkhhkxty.1. This dataset is distributed under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence allowing free use and modification with proper attribution to the original creators.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Muhammad Fazle Mawla: Conceptualization, Methodology, Resources, Writing, Review & Editing. **Julian Alby Nathanael:** Conceptualization, Methodology, Resources, Writing, Review & Editing. **Stefanus Benhard:** Validation, Supervision. **Jeffrey Junior Tedjasulaksana:** Validation, Supervision.

AI USAGE DECLARATION

We declare that we used AI tools, specifically Google Gemini with the specific purposes to coherently describe the manuscript structure, check for grammar mistakes, and paraphrase sentences. All output generated by AI were carefully reviewed, edited, and paraphrased with careful verification by the authors to ensure clarity and academic integrity.

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